



Seasonal Hydrogeochemical and Geospatial Assessment of Agro-Geogenic Groundwater Contamination in Mining-Influenced Terrain, SW Y.S.R. Kadapa District, A.P., India

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Abstract A total of 100 groundwater samples collected during the pre-monsoon (PRM) and post-monsoon (POM) seasons of 2019 were analyzed to evaluate hydrogeochemistry and groundwater quality in the southwestern part of Y.S.R. Kadapa District, Andhra Pradesh, India. Groundwater is predominantly alkaline and hard in nature. Nitrate concentrations exceeded the drinking water limit (45 mg/L) in 42% of samples during both seasons, while fluoride exceeded permissible limits in 32% (PRM) and 26% (POM) of samples. Hydrochemical facies are dominated by Ca–HCO₃ (32% PRM; 38% POM) and mixed Ca–Mg–Cl (26% PRM; 28% POM) types, indicating progressive geochemical evolution. Gibbs plot analysis indicates rock dominance as the primary controlling mechanism. Correlation analysis

reveals a strong positive association between Mg²⁺ and NO₃[−] ($r=0.81$ PRM; $r=0.76$ POM) and between EC and TDS ($r=0.91$ PRM; $r=0.78$ POM). Factor analysis extracted five principal components explaining 74.54% (PRM) and 71.33% (POM) of the total variance, reflecting combined geogenic and anthropogenic influences. The statistical results indicate significant nitrate and fluoride contamination, rendering a substantial portion of groundwater unsuitable for drinking. Water Quality Index (WQI) results indicate that 26% (PRM) and 22% (POM) of samples fall under poor to very poor categories, whereas Total Hazard Index (THI) values exceed unity in 38% of locations, particularly affecting children and infants. Geospatial mapping using IDW interpolation delineates distinct contamination hotspots associated with agricultural zones and structurally controlled lithological units. Overall, groundwater quality is predominantly governed by agro-geogenic processes with localized anthropogenic and minor mining impacts. The combined geogenic and anthropogenic impacts render a considerable portion of groundwater unsuitable for direct drinking purposes, emphasizing the need for continuous monitoring and targeted management strategies.

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1 Introduction

Groundwater is the principal source of drinking water in semi-arid regions of India, particularly in hard-rock terrains such as the Y.S.R Kadapa District, where it occurs mainly in weathered and fractured zones. In such settings, groundwater chemistry is controlled by complex interactions among lithology, mineral weathering, ion exchange, recharge dynamics, and anthropogenic inputs (Awasthi et al., 2026; Karunanidhi et al., 2019; Narasimha & Sudarshan, 2017; Wu & Sun, 2016). The southwestern part of Y.S.R. Kadapa district is characterized by carbonate formations, shales, quartzites, and igneous intrusions that significantly influence hydrogeochemical evolution. Previous investigations in different mandals of the Kadapa Basin have consistently identified silicate weathering, carbonate dissolution, and water–rock interaction as the dominant mechanisms governing groundwater chemistry (Suvarna et al., 2022; Sunitha et al., 2026), largely attributing groundwater quality deterioration to natural geogenic processes.

Despite these extensive studies, a critical knowledge gap remains. Most, earlier research has treated groundwater systems in the Kadapa District as predominantly rock-dominated and has not explicitly evaluated the hydrogeochemical influence of inactive and abandoned mining areas (Shinde et al., 2024). The southwestern part of Y.S.R. Kadapa district hosts several inactive limestone and barite mines, yet no study has systematically assessed groundwater quality in direct spatial relation to these sites. Moreover, limited attention has been given to seasonal variability in contaminant mobilization from mine spoils, and few investigations have integrated geospatial modeling with multivariate statistical tools to distinguish mining-related impacts from natural geogenic controls (Sharma et al., 2026; Sudharsan Reddy et al., 2024; Suvarna et al., 2022).

Inactive mining areas require separate and focused assessment because, after the cessation of operations, mine pits, waste dumps, exposed rock surfaces, and abandoned shafts remain unmanaged. During monsoonal recharge, rainwater interacts with these materials, enhancing the leaching of dissolved ions and potentially toxic constituents into shallow aquifers through fractures and faults (Rana et al., 2026; Sanjupriya et al., 2025). Such processes can create localized contamination hotspots that are not evident when

groundwater chemistry is interpreted solely through regional lithological controls (Jena et al., 2025). Numerous studies have reported adverse environmental and health impacts associated with abandoned mining sites (Patel et al., 2026), emphasizing the need for site-specific hydrogeochemical evaluation.

Geospatial analysis plays a critical role in strengthening such assessments by integrating hydrochemical data with spatial proximity around mining sites. GIS-based spatial interpolation and thematic mapping enable the identification of contamination gradients, hotspot zones, and seasonal shifts in groundwater quality at finer spatial resolution (Bhatnagar & Thakral, 2024; Maurya et al., 2025; Sudharsan Reddy et al., 2024;). When combined with multivariate statistical techniques, geospatial tools improve the differentiation of localized anthropogenic influences from broader lithology-controlled processes (Gajbhiye et al., 2016; Sudharshan Reddy et al., 2022; Suvarna et al., 2022), thereby enhancing the reliability of groundwater contamination assessment in complex hard-rock terrains.

To address the identified gaps, the present study was specifically designed to evaluate groundwater quality within a 5 km radius of inactive mining sites in the southwestern Y.S.R. Kadapa district through systematic pre- and post-monsoon sampling. By integrating hydrogeochemical facies analysis, seasonal comparison, multivariate statistical techniques, and GIS-based spatial modeling, this study provides new insight into spatial contamination dynamics, seasonal recharge-driven mobilization, and statistical differentiation of mining-related impacts from natural geogenic processes. The findings contribute to a more refined understanding of groundwater quality evolution in mining-affected regions of the Kadapa Basin and support the development of targeted monitoring and management strategies.

2 Study Area

The study area is situated in the southwestern part of Kadapa District and lies between 14°21'00" N latitude and 78°22'30" E longitude, covering an area of approximately 900 km². The region falls within the Survey of India topographic sheets 57 J/03, 57 J/06, and 57 J/07 at a scale of 1:50,000. Administratively, the area comprises four revenue mandals i.e.

Vempalli, Vemula, Pulivendla, and Lingala as illustrated in Fig. 1.

The study region is of considerable economic significance due to the occurrence of various mineral resources and construction materials. Previous studies by Murthy (1981) and Kasipathi et al. (2008) reported widespread deposits of barite, asbestos, limestone, iron, copper, lead–zinc, and talc across the area. From a hydrogeological standpoint, groundwater availability in this drought-prone region is predominantly controlled by basin rocks, particularly

carbonate formations and shales, which act as the primary aquifer systems. The mandal-wise distribution of different lithological units in the southwestern part of Kadapa District is presented in Table 1.

2.1 Sampling Procedure

The present study aimed to assess groundwater quality in areas surrounding inactive mine sites in the southwestern part of Y.S.R. Kadapa District, South India. A total of 100 groundwater samples

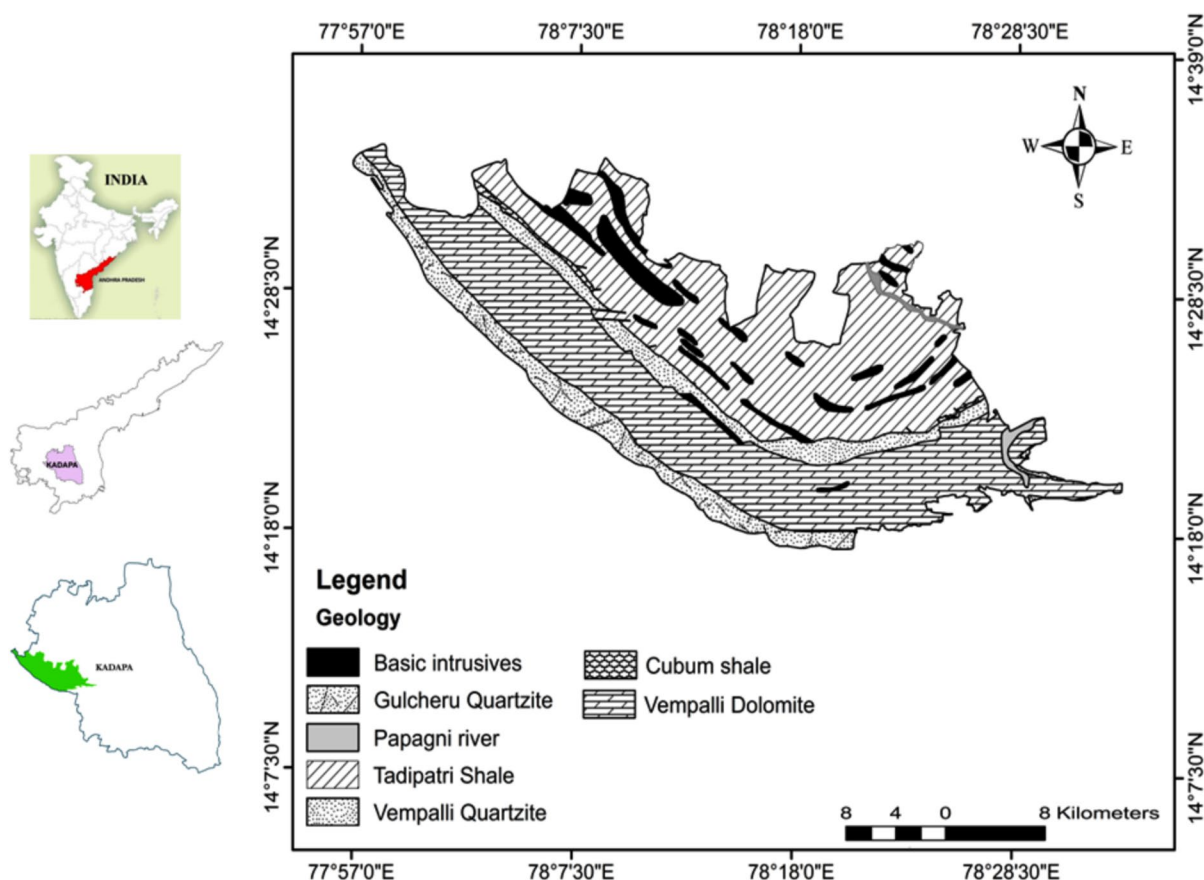


Fig. 1 Location map of the study area demarcated with exposure of local lithological units

Table 1 Mandal wise distributions of Litho units in SW region of the Y.S.R Kadapa district

Sl. No	Name of the Mandal	Rock formation
1	Vempalli	Dolomitic rocks, Quartzites, Volcanic flows, shale
2	Vemula	
3	Pulivendula	
4	Lingala	

were collected from bore wells within a 5 km radius of the inactive mining areas during the PRM and POM seasons of 2019, considering the fractured hard rock hydrogeology of Y.S.R. Kadapa District where groundwater can migrate several kilometers through interconnected fractures. This buffer ensured adequate spatial coverage to capture the hydrochemical influence of inactive mines, and sampling locations were selected based on bore well availability (Fig. 2).

To ensure spatial accuracy and consistency, the sampling points were precisely identified using a Global Positioning System (GPS). Groundwater samples were collected in pre-cleaned and air-dried polyethylene bottles, preserved at approximately 4 °C using ice packs, and transported to the laboratory under dark conditions to prevent chemical alteration. All samples were analyzed within 72 h of collection

at the C-MET laboratory, Hyderabad, Telangana State, India.

3 Materials and Methods

3.1 Analysis of Physico-Chemical Parameters

During 2019, a total of 100 groundwater samples were collected from 50 villages located near inactive mining areas during both the pre-monsoon (PRM) and post-monsoon (POM) seasons. Bore well depths ranged from 20 to 60 m below ground level. Sampling locations were recorded using GPS to ensure spatial accuracy. Samples were collected in 2 L pre-cleaned polyethylene bottles, properly labeled, and preserved following standard protocols. The same methodology was adopted in both seasons. Physicochemical

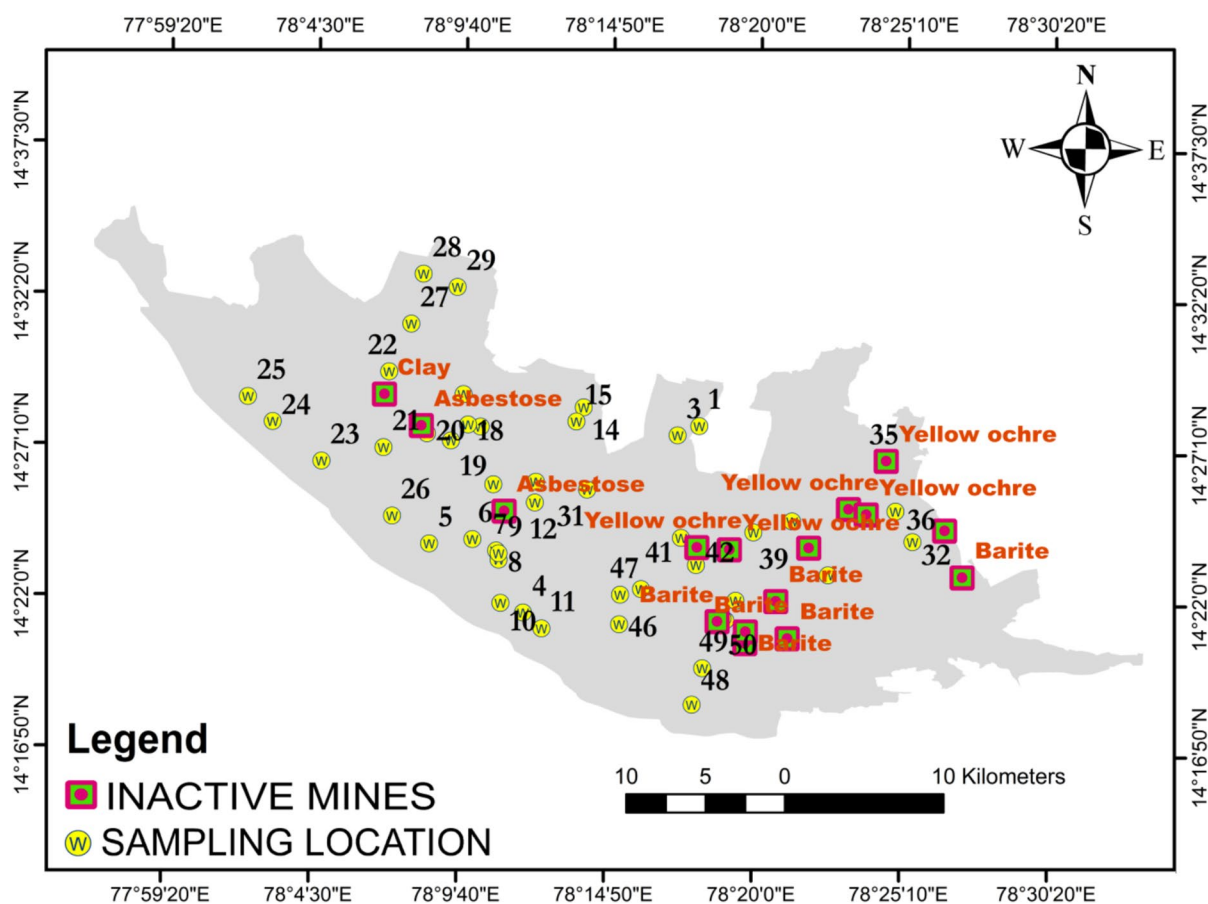


Fig. 2 Map showing the locations of water samples and their proximity to inactive mining sites

parameters were analyzed as per Standard Methods for the Examination of Water and Wastewater (APHA, 2023). Parameters included pH, EC, TDS, TH, Ca^{2+} , Mg^{2+} , Na^+ , K^+ , CO_3^{2-} , HCO_3^- , Cl^- , SO_4^{2-} , NO_3^- , and F^- . Field parameters (pH, EC, TDS) were measured using a Water Analyzer 371. Major cations and anions were determined through titrimetric, flame photometric (Systronics Model 128), spectrophotometric, and ion-selective electrode (Orion 4-Star Ion Meter) methods. In addition, heavy metals such as As, Cd, Co, Cr, Fe, Li, Mn, Ni, Pb, Sr, and Zn were analyzed using standard instrumental techniques (ICP-OES/Atomic Absorption Spectrophotometer) at C-MET Laboratory, Hyderabad. Ionic Balance Error ($\pm 5\%$) confirmed analytical precision and data reliability. The overall methodology adopted in the study is illustrated in Fig. 3.

3.2 Multivariate Statistical Analysis

Multivariate statistical techniques were applied to better understand the relationships among the groundwater quality parameters and to identify the major factors influencing groundwater chemistry. Pearson correlation matrix, Hierarchical Cluster Analysis (HCA), and Factor Analysis were performed using the Statistical Package for Social Sciences (SPSS – version 64). These techniques help in simplifying complex hydrochemical data and in distinguishing between natural and anthropogenic influences (Etikala et al., 2024).

3.3 Correlation Matrix

Pearson's correlation analysis was used to determine the degree of relationship between different physico-chemical parameters. The correlation coefficient indicates both the strength and direction of the linear relationship between two variables. A strong positive correlation suggests that the parameters may have a common source or are influenced by similar geochemical processes, whereas a negative correlation indicates inverse behavior. This analysis helps in identifying possible associations such as mineral dissolution, ion exchange, or external contamination sources (Raju & Singh, 2017).

3.4 Factor Analysis

Factor analysis was carried out to identify the underlying processes controlling groundwater chemistry. This statistical method reduces a large number of variables into a smaller set of factors without losing significant information. In the present study, R-mode factor analysis was applied, which focuses on relationships among variables to detect common controlling factors. The extracted factors represent possible hydrogeochemical processes such as rock–water interaction, agricultural return flow, or mining influence. Factor analysis is widely used in geochemical studies to interpret complex datasets and to understand aquifer behavior (Yonas et al., 2026).

3.5 Geospatial Modeling

Geospatial modelling of groundwater quality was carried out using ArcGIS 10.8 and ERDAS IMAGINE 2014 by integrating field-based hydrochemical data with spatial datasets in a GIS environment. The Aspect map was prepared from ASTER DEM data with 30 m spatial resolution using the Spatial Analyst tools to understand slope orientation and its influence on groundwater movement and recharge. Slope and elevation maps were also generated from the same DEM at 30 m raster resolution. Land Cover map for the year 2019 were prepared using IRS-P6 LISS-III imagery (30 m resolution) through supervised classification in ERDAS IMAGINE, followed by editing and accuracy checking in ArcGIS; all layers were resampled to 30 m resolution for uniform analysis. Groundwater flow direction was derived from the DEM using Flow Direction and Flow Accumulation tools based on the D8 algorithm to identify flow paths and potential contaminant transport zones. Spatial distribution maps of groundwater quality parameters were generated using the Inverse Distance Weighted (IDW) interpolation method in ArcGIS 10.8 with an output cell size of 30 m. Cross-validation was carried out and finally, all thematic layers were projected to a common coordinate system and integrated in a raster-based GIS framework to delineate groundwater contamination zones and analyze their spatial relationship with terrain and land use characteristics.

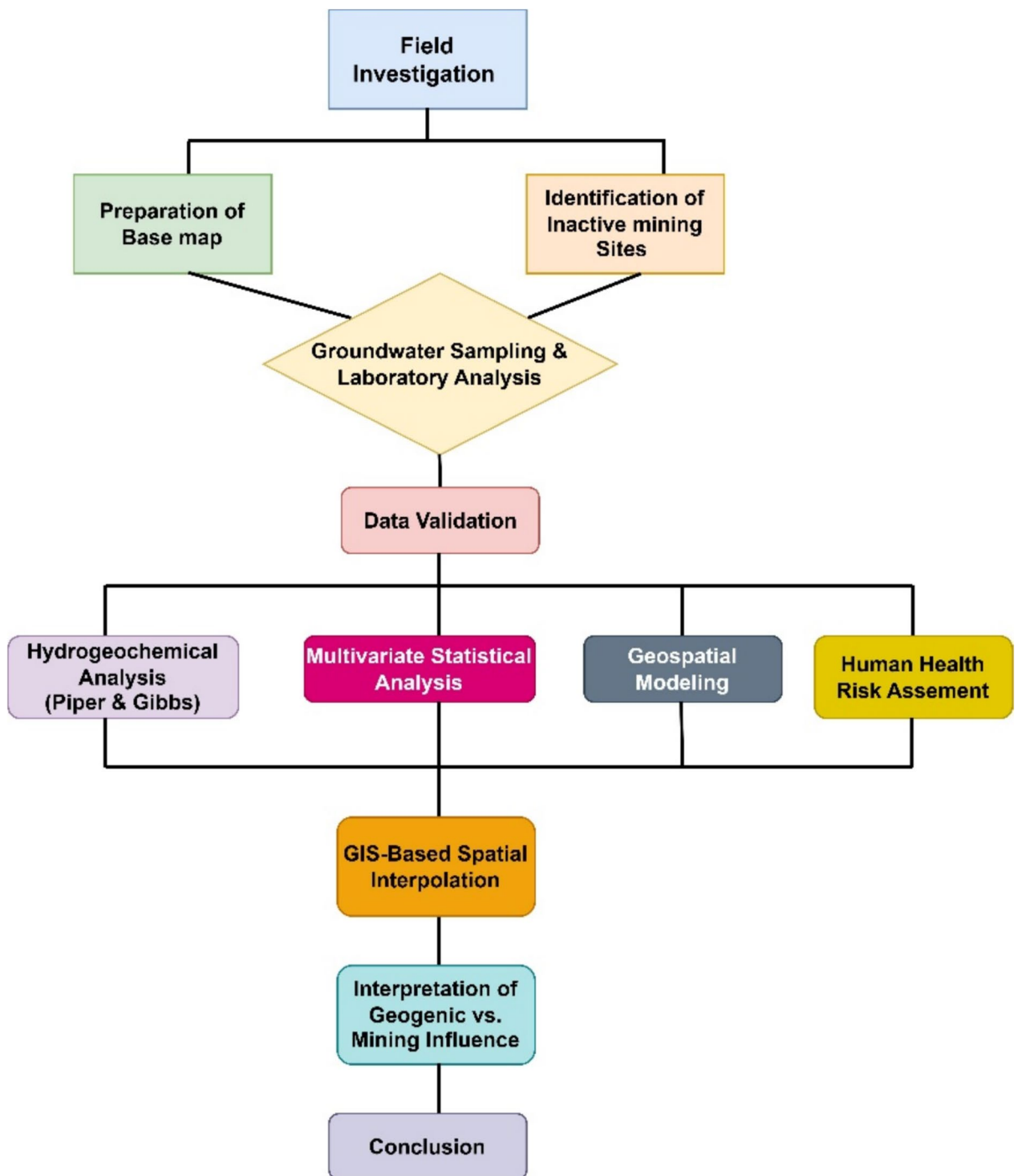


Fig. 3 Comprehensive Methodological Framework for Groundwater Quality Assessment for drinking purpose

3.6 Water Quality Index (WQI) Estimation

The overall suitability of groundwater for drinking purposes was evaluated using the Water Quality Index (WQI) based on the weighted arithmetic index method developed by Usman et al., 2024. The calculation of WQI was carried out in three main steps.

Step 1: Calculation of Quality Rating (qn)

$$q_n = \left[\frac{V_n - V_i}{S_n - V_i} \right] \times 100$$

The quality rating for each parameter was calculated using the following equation:

where:

q_n quality rating of the nth parameter.

V_n observed concentration of the nth parameter.

S_n recommended standard value of the parameter.

V_i ideal value of the parameter.

For drinking water assessment, the ideal value (V_i) was taken as zero for all parameters except pH, for which the ideal value was considered as 7.0.

Step 2: Calculation of Unit Weight (W_n)

$$W_n = \frac{K}{S_n}$$

Each parameter was assigned a unit weight inversely proportional to its recommended standard value using the formula:

where:

W_n unit weight of the nth parameter.

S_n standard permissible value.

K proportionality constant.

The constant (K) was calculated as:

$$k = \frac{1}{\sum \frac{1}{S_n}}$$

This approach ensures that parameters with lower permissible limits (more critical to health) are given higher weight.

Step 3: Calculation of Overall WQI.

The final WQI was calculated by combining the quality ratings and their respective unit weights using the expression:

$$WQI = \frac{\sum q_n W_n}{\sum W_n}$$

The computed WQI values were then classified into different categories such as excellent, good, poor, very poor, and unsuitable for drinking, providing a simplified representation of overall groundwater quality.

3.7 Health Risk Assessment (HHRA)

Human Health Risk Assessment (HHRA) was carried out to evaluate the potential non-carcinogenic risks associated with fluoride (F^-) and nitrate (NO_3^-) in groundwater. The assessment followed the four-step approach recommended by the United States Environmental Protection Agency (USEPA): risk identification, dose assessment, exposure assessment, and hazard characterization (Ullengula et al., 2026). In this study, fluoride and nitrate were selected due to their significant health impacts when present beyond permissible limits (USEPA, 2008).

The average daily dose (ADD) through ingestion was calculated using the following equation:

$$ADD = \frac{C_{PW} \times IR \times ED \times EF}{ABW \times AET}$$

where C_{PW} is the concentration of fluoride or nitrate in groundwater (mg/L), IR is the ingestion rate (2.5 L/day for adults, 0.78 L/day for children, and 0.3 L/day for infants), ED is exposure duration (64 years for adults, 12 years for children, and less than 1 year for infants), EF is exposure frequency (365 days/year), ABW is average body weight (57.5 kg for adults, 18.7 kg for children, and 16.9 kg for infants), and AET is average exposure time in days (USEPA, 2011).

The non-carcinogenic risk was evaluated using the Hazard Quotient (HQ), calculated as:

$$HQ = \frac{ADD}{RfD}$$

where RfD is the reference dose (0.04 mg/kg/day for fluoride and 1.6 mg/kg/day for nitrate). If the HQ value is less than 1, no significant non-carcinogenic risk is expected. However, an HQ value greater than 1 indicates a potential health risk (USEPA, 2012).

The Total Hazard Index (THI), which represents the combined non-carcinogenic risk from multiple contaminants, was calculated as:

$$THI = \sum HQ_i$$

A THI value greater than 1 suggests a possible overall health risk due to combined exposure. This assessment provides a clear understanding of the potential health implications of consuming groundwater in the study area, particularly for vulnerable groups such as children and infants.

4 Results and Discussion

4.1 Hydrogeochemical Characteristics of Groundwater

The statistical summary of physicochemical parameters & comparison of ion concentrations during the pre-monsoon (PRM) and post-monsoon (POM) seasons with the drinking water standards prescribed by the World Health Organization (WHO, 2011) is presented in Table 2.

4.2 pH

Groundwater pH values ranged from 7.4 to 8.9 during the PRM season and from 7.2 to 8.6 during the POM season of 2019, indicating a slightly alkaline nature. A marginal decrease in pH during the post-monsoon period may be attributed to reduced CO₂ availability in groundwater following dilution effects (Gyanendra & Alam, 2023). The conversion of bicarbonate (HCO₃⁻) to carbonate (CO₃²⁻) was minimal, as such reactions predominantly occur at higher pH levels (> 8). Consequently, carbonate precipitation and associated calcium carbonate (CaCO₃) formation were limited. Overall, bicarbonate and carbonate ions are

the principal contributors controlling groundwater pH in the study area.

4.3 Electrical Conductivity (EC) and Total Dissolved Solids (TDS)

Electrical conductivity, which reflects the total ionic concentration of groundwater, varied from 115 to 2920 µS/cm during PRM and from 110 to 2800 µS/cm during POM. Higher EC values observed during the PRM season suggest increased ionic enrichment prior to monsoonal dilution. The wide range and high standard deviation indicate significant spatial variability, likely influenced by geochemical processes, anthropogenic inputs, and excessive groundwater withdrawal (Subba Rao et al., 2017; Narasimha & Sudharshan, 2017).

Total dissolved solids ranged from 110 to 1987 mg/L in PRM and from 90 to 1820 mg/L in POM. Elevated (2%) TDS levels in both seasons, are mainly attributed to salt leaching, agricultural practices, and mining-related activities. TDS plays a crucial role in assessing water quality, as it directly influences water hardness and overall potability.

4.4 Total Hardness (TH)

Total hardness values ranged from 75 to 530 mg/L during PRM and from 70 to 520 mg/L during POM. The pronounced spatial variability in hardness across both seasons may be linked to the dissolution of calcium and magnesium-bearing minerals, industrial effluents, untreated sewage discharge, and recharge-induced leaching processes (Madhusudhan et al., 2024).

4.5 Major Cations

Sodium concentrations varied from 11.2 to 429.5 mg/L during PRM and from 12.4 to 430 mg/L during POM. Based on WHO (2011) guidelines, 6% of PRM samples and 4% of POM samples exceeded the permissible limit of 200 mg/L. Elevated sodium levels can adversely affect soil permeability and contribute to alkaline soil conditions. These variations are primarily attributed to mineral weathering (plagioclase feldspars, amphiboles, and clay minerals), leaching processes, and base-exchange reactions involving clay and zeolite minerals.

Table 2 Statistical summary of physico-chemical parameters and comparison with world health organization (WHO) drinking water standards for groundwater during PRM and POM seasons – 2019 (n = 100)

Chemical parameter	Expressed	PRM-2019			POM-2019			Ranges of standards			W.H.O limits in PRM-2019	W.H.O limits in POM-2019
		Minimum	Maximum	Mean	Minimum	Maximum	Mean	W.H.O (Desirable—Permissible)	Mean	Maximum		
pH	Units	7.4	8.9	8.2	7.2	8.6	8	7–9.2	8	26	26	6
EC	$\mu\text{S}/\text{cm}$	115	2920	1261	110	2800	1169.6	–	1169.6	–	–	–
TDS	mg/L	110	1987	677	90	1820	658.7	500–1500	658.7	2	2	2
TH	mg/L	75	530	262	70	520	245.5	100–500	245.5	4	4	2
Na ⁺	mg/L	11.2	429.5	85.4	12.4	430	87.6	50–200	87.6	6	6	4
K ⁺	mg/L	0.5	37.5	6.1	0.3	32.2	5.9	10	5.9	12	12	12
Ca ²⁺	mg/L	6.9	103.2	47	7.3	110	49.4	75–200	49.4	Nil	Nil	Nil
Mg ²⁺	mg/L	11.6	152.5	52.5	12	162.24	55.4	30–150	55.4	2	2	4
CO ₃ ²⁻	mg/L	0	54	22.2	0	52	20.9	–	20.9	–	–	–
HCO ₃ ⁻	mg/L	28	340.2	191.2	22	320	172.6	300–600	172.6	Nil	Nil	Nil
SO ₄ ²⁻	mg/L	10.7	802.2	89.7	9.43	220.32	69.4	200–600	69.4	2	2	Nil
Cl ⁻	mg/L	23.9	322	82.5	0	320	74.8	250–600	74.8	Nil	Nil	Nil
NO ₃ ⁻	mg/L	0.1	589	78.1	0.1	560	74.7	45	74.7	42	42	42
F ⁻	mg/L	0.4	4.7	1.5	0.4	4.5	1.4	0.5–1.5	1.4	32	32	26

Potassium concentrations ranged from 0.5 to 37.5 mg/L during PRM and from 0.3 to 32.2 mg/L during POM. Approximately 12% of samples in both seasons exceeded the WHO permissible limit of 10 mg/L. The primary sources of potassium include weathering of potash-bearing silicate minerals, fertilizer application, rainfall infiltration, and irrigation return flow. Its relatively lower mobility compared to sodium is due to adsorption onto clay minerals and resistance to chemical weathering.

Calcium concentrations ranged from 6.9 to 103.2 mg/L during PRM and from 7.3 to 110 mg/L during POM. All samples were within the permissible limit of 200 mg/L (WHO, 2011). Elevated calcium levels are mainly controlled by chemical weathering and dissolution of carbonate rocks in the study area (Sudhakar & Narsimha, 2013).

Magnesium concentrations varied from 11.6 to 152.5 mg/L in PRM and from 12.0 to 162.2 mg/L in POM. About 8% of samples in both seasons exceeded the WHO limit of 150 mg/L. High magnesium levels are associated with the dissolution of dolomite, limestone, gypsum, anhydrite, and clay-rich sediments during groundwater recharge (Sreelesh et al., 2025).

4.6 Major Anions

Bicarbonate was the dominant anion, with concentrations ranging from 28 to 340 mg/L in PRM and from 22 to 320 mg/L in POM. The observed spatial variation is primarily governed by chemical weathering processes (Magesh & Chandrasekar, 2013). Most samples remained within the permissible limit of 300 mg/L prescribed by WHO and BIS.

Sulfate concentrations ranged from 10.7 to 802 mg/L during PRM and from 9.4 to 220 mg/L during POM. Only 2% of PRM samples exceeded the WHO guideline value of 600 mg/L. Sulfate enrichment is mainly derived from the dissolution of gypsum, anhydrite, and pyrite-bearing formations, as well as agricultural inputs and organic matter degradation (Brindha & Elango, 2013).

Chloride levels varied between 23.9 and 322 mg/L in PRM and from negligible values to 330 mg/L in POM. Elevated chloride concentrations often serve as indicators of anthropogenic pollution and groundwater contamination. Possible sources include rock weathering, soil leaching, and domestic and municipal effluents (Subba Rao et al., 2017).

Nitrate concentrations showed a wide range, from 0.1 to 589 mg/L during PRM and from 0.1 to 560 mg/L during POM. Approximately 42% of samples in both seasons exceeded the WHO permissible limit of 45 mg/L. High nitrate levels are primarily attributed to agricultural runoff, excessive fertilizer application, sewage leakage, and waste disposal in abandoned mining areas (Panaskar et al., 2016; Muralidhara Reddy et al., 2019). Intensive fertilizer use for rice cultivation further exacerbates nitrate enrichment in the region (CGWB, 2016).

Fluoride concentrations ranged from 0.4 to 4.7 mg/L during PRM and from 0.4 to 4.5 mg/L during POM. Based on WHO guidelines, 32% of PRM samples and 26% of POM samples exceeded the permissible limit of 1.5 mg/L. Fluoride enrichment is mainly of geogenic origin, linked to water–rock interaction, ion exchange processes, and dissolution of fluoride-bearing minerals in limestone, shale, and carbonate formations (Brindha et al., 2011; Raju et al., 2011). Secondary contributions from phosphate fertilizer usage are also evident (Subba Rao et al., 2017).

4.7 Water Quality Index Assessment

From the WQI classification (Table 3) for both pre-monsoon (PRM) and post-monsoon (POM) seasons indicate that the majority of groundwater samples fall under the “Good” category (Class II: 51–100). During the Pre-Monsoon season, 34 samples (68%) were classified as Good, while 12 samples (24%) fell under Poor Water (Class III: 101–200). A smaller proportion, 3 samples (6%) were categorized as Excellent (Class I: < 50). Only 1 sample (2%) was identified as Very Poor Water (Class IV: 201–300), and no samples were recorded in Class V (> 300), indicating that none were unsuitable for drinking.

Similarly, in the post-monsoon season, 34 samples (68%) remained in the good category. The proportion of Poor Water samples slightly decreased to 10 samples (20%), while Excellent water samples increased to 5 (10%). One sample (2%) continued to fall under the Very Poor Water category.

Over all Water Quality Index (WQI) results indicate that groundwater samples in the poor to very poor-quality categories constitute 26% during the pre-monsoon season (24% poor and 2% very poor) and 22% during the post-monsoon season (20% poor and

Table 3 WQI at individual sampling stations for PRM and POM season in 2019

Class	WQI Value	Water Quality Status	2019			
			PRM		POM	
			No of samples	% of samples	No of samples	% of samples
I	< 50	Excellent	3	6	5	10
II	51–100	Good	34	68	34	68
III	101–200	Poor Water	12	24	10	20
IV	201–300	Very Poor Water	1	2	1	2
V	> 300	Water Un Suitable for Drinking	NIL	NIL	NIL	NIL

2% very poor). No samples were classified as unsuitable for drinking in this season as well.

The predominance of “Good” category water in both seasons indicates that groundwater chemistry is mainly governed by natural processes such as rock–water interaction, mineral dissolution, and ion exchange. The increase in “Excellent” samples during the post-monsoon season reflects dilution due to rainfall recharge, which lowers dissolved ion concentrations. However, the occurrence of “Poor” and “Very Poor” samples, particularly near inactive mining areas, may result from leaching of mine spoils, oxidation of exposed minerals, and mobilization of dissolved solids. Seasonal evaporation during pre-monsoon further concentrates ions. Overall, groundwater quality is generally acceptable, with localized contamination requiring monitoring.

4.8 Hydrogeochemical Facies and Controlling Mechanisms

The relative abundance of major ions followed the order $\text{Na}^+ > \text{Mg}^{2+} > \text{Ca}^{2+} > \text{K}^+$ for cations and $\text{HCO}_3^- > \text{Cl}^- > \text{SO}_4^{2-} > \text{NO}_3^- > \text{CO}_3^{2-} > \text{F}^-$ for anions during both seasons. Based on this composition, groundwater in the study area predominantly belongs to the Ca–Mg–Cl and Ca–HCO₃ types.

4.8.1 Piper Diagram Interpretation

Piper’s trilinear diagram (Piper, 1953) was employed to identify groundwater facies and geochemical evolution. The diagram consists of cation and anion triangles and a central diamond field, enabling classification based on ionic dominance. The groundwater samples were categorized into six facies: Ca–HCO₃,

Na–Cl, mixed Ca–Na–HCO₃, mixed Ca–Mg–Cl, Ca–Cl, and Na–HCO₃.

From the Fig. 4, it indicates that 32% (PRM) and 38% (POM) of samples belonged to the Ca–HCO₃ facies, representing recharge zones. Mixed Ca–Mg–Cl facies accounted for 26% (PRM) and 28% (POM), indicating ion exchange processes associated with clay-rich formations. Na–Cl and Ca–Cl facies reflect secondary salinization and leaching of salts. Overall, groundwater evolution in the study area follows a transition from Ca–HCO₃ to Ca–Cl and finally to mixed Ca–Mg–Cl facies, highlighting the influence of water–rock interaction, ion exchange, and anthropogenic inputs.

4.9 Gibbs Diagram Interpretation

The Gibbs diagram (Gibbs, 1970) was employed to evaluate the processes controlling groundwater chemistry in the study area. This diagram is widely used to identify the dominant mechanisms influencing groundwater composition, such as precipitation, rock–water interaction, and evaporation. In the present study, the ratios of major anions (Cl^-)/($\text{Cl}^- + \text{HCO}_3^-$) and cations ($\text{Na}^+ + \text{K}^+$)/($\text{Na}^+ + \text{K}^+ + \text{Ca}^{2+}$) were plotted against total dissolved solids (TDS) to construct the Gibbs plots. The cation ratio (Gibbs ratio 1) and anion ratio (Gibbs ratio 2) were calculated using the following expressions:

$$\text{Gibbsratio1} = (\text{Cl}^-) / (\text{Cl}^- + \text{HCO}_3^-)$$

$$\text{Gibbsratio1} = (\text{Na}^+ + \text{K}^+) / (\text{Na}^+ + \text{K}^+ + \text{Ca}^{2+})$$

From the Fig. 5, it is evident that most groundwater samples in both pre-monsoon and post-monsoon seasons fall within the rock dominance field of the

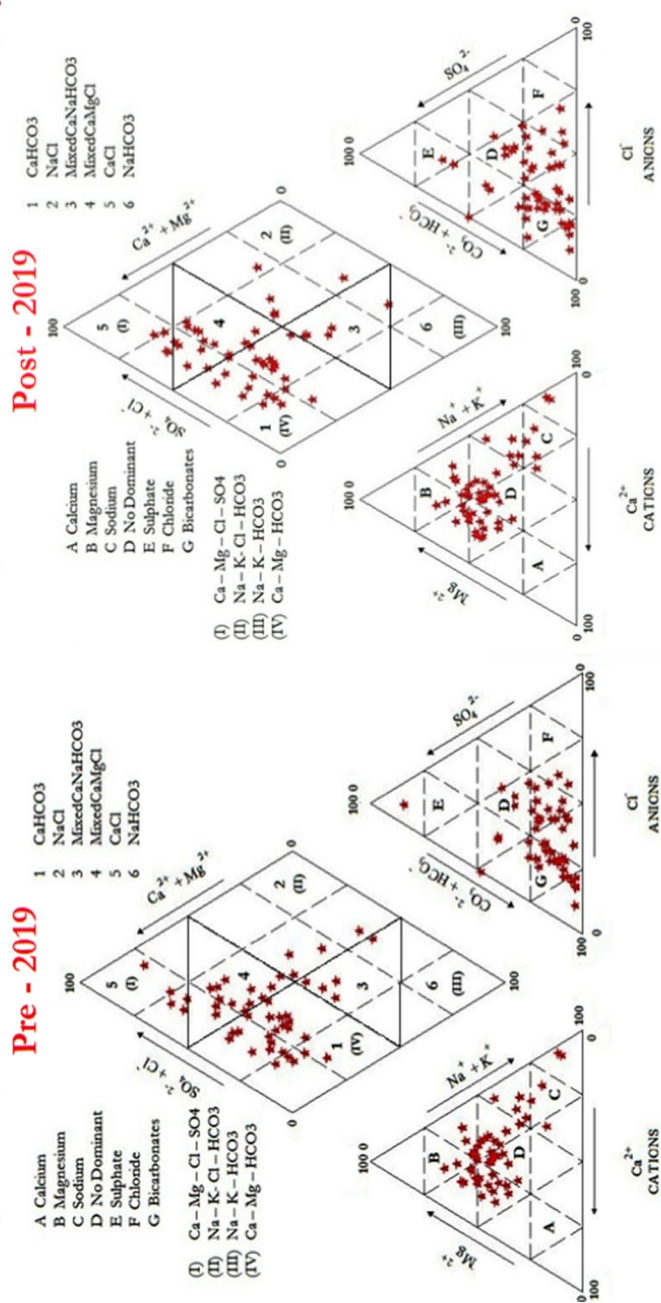


Fig. 4 Piper diagram showing water type in study area **A** PRM season **B** POM season

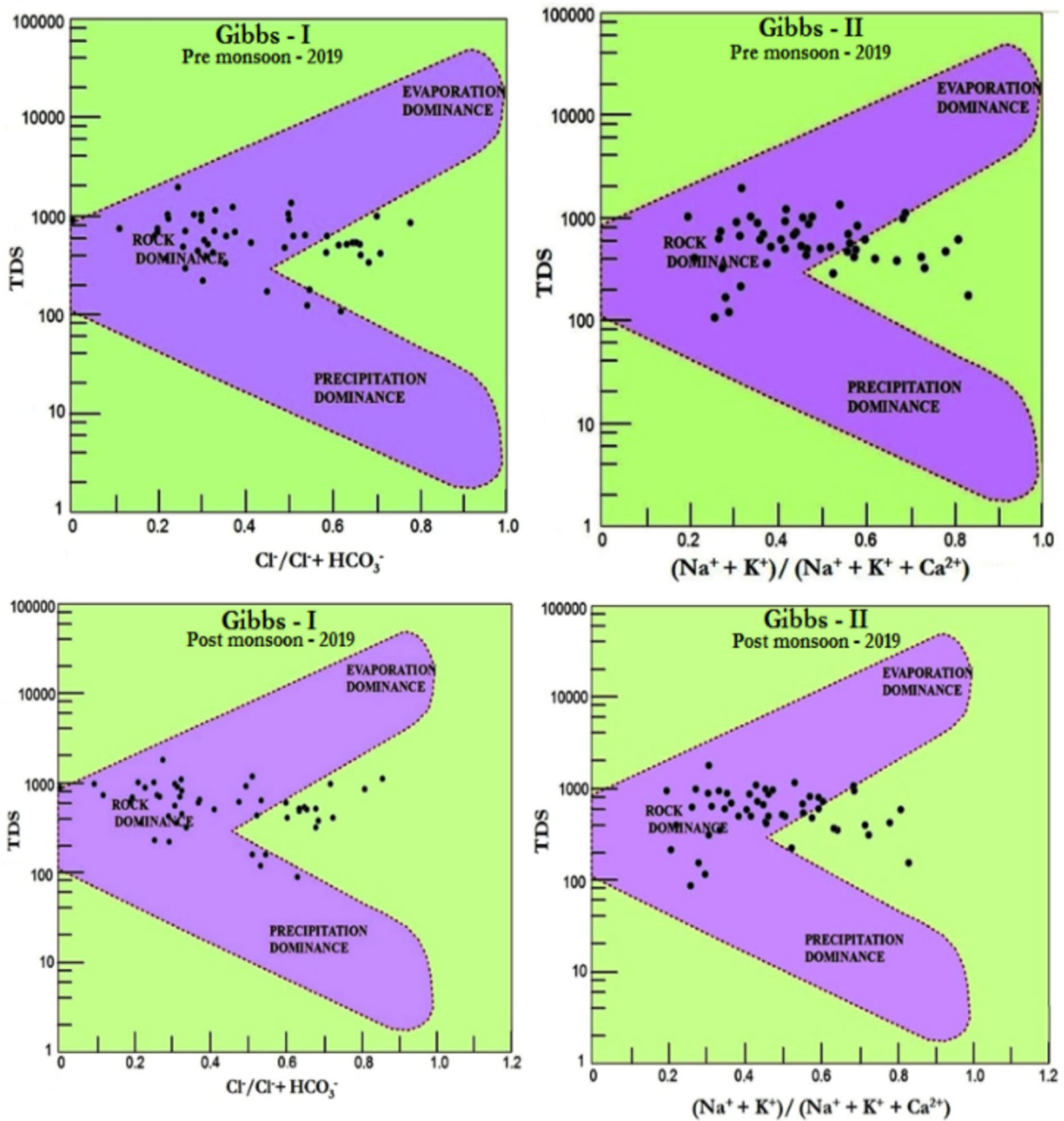


Fig. 5 Gibbs (1970) plots showing the controlling chemistry mechanism of groundwater during PRM and POM season in 2019

Gibbs diagram. This indicates that groundwater chemistry is mainly controlled by water–rock interaction processes. The dissolution of minerals, silicate weathering, and ion exchange between groundwater and aquifer materials play a major role in controlling the chemical composition. A few samples plot toward the precipitation dominance field, particularly during

the post-monsoon season, indicating the influence of rainfall recharge and dilution of dissolved ions. Some samples also show a slight shift toward the evaporation dominance field, especially those with higher TDS values, suggesting that evaporation during dry periods increases the concentration of dissolved salts. The enrichment of Na^+ , Cl^- , and TDS in certain

locations reflects the combined effects of natural weathering, evaporation, irrigation return flow, and other human activities. Overall, the Gibbs diagrams confirm that natural geological processes are the dominant factors controlling groundwater chemistry, with minor seasonal and anthropogenic influences.

4.9.1 Hydrogeochemical Evolution

Understanding the hydrogeochemical evolution of the study area requires evaluating the combined effects of lithology, mineral weathering, ion exchange, and anthropogenic inputs. Based on the geochemical evidence, the controlling mechanisms can be clearly prioritized.

4.10 Primary Mechanism – Silicate Weathering

The bivariate plots (Fig. 6) of $\text{Ca}^{2+}/\text{Na}^+$ versus $\text{HCO}_3^-/\text{Na}^+$ and $\text{Ca}^{2+}/\text{Na}^+$ versus $\text{Mg}^{2+}/\text{Na}^+$, together with the dominance of samples in the rock weathering field of

the Gibbs diagram, indicate that groundwater chemistry is mainly governed by silicate weathering. The release of Na^+ and K^+ and the observed Mg^{2+} – HCO_3^- relationships further support rock–water interaction as the principal source of dissolved ions.

4.11 Secondary Mechanism – Cation Exchange and Limited Carbonate Dissolution

The plot of $(\text{Ca}^{2+} + \text{Mg}^{2+})$ versus $(\text{HCO}_3^- + \text{SO}_4^{2-})$ (Fig. 7) shows most samples below the equiline, indicating active cation exchange during groundwater movement. Carbonate dissolution contributes locally but is not the dominant process.

4.12 Minor Mechanism – Evaporation and Anthropogenic Inputs

A few samples show slight enrichment in dissolved ions, suggesting minor evaporation effects and

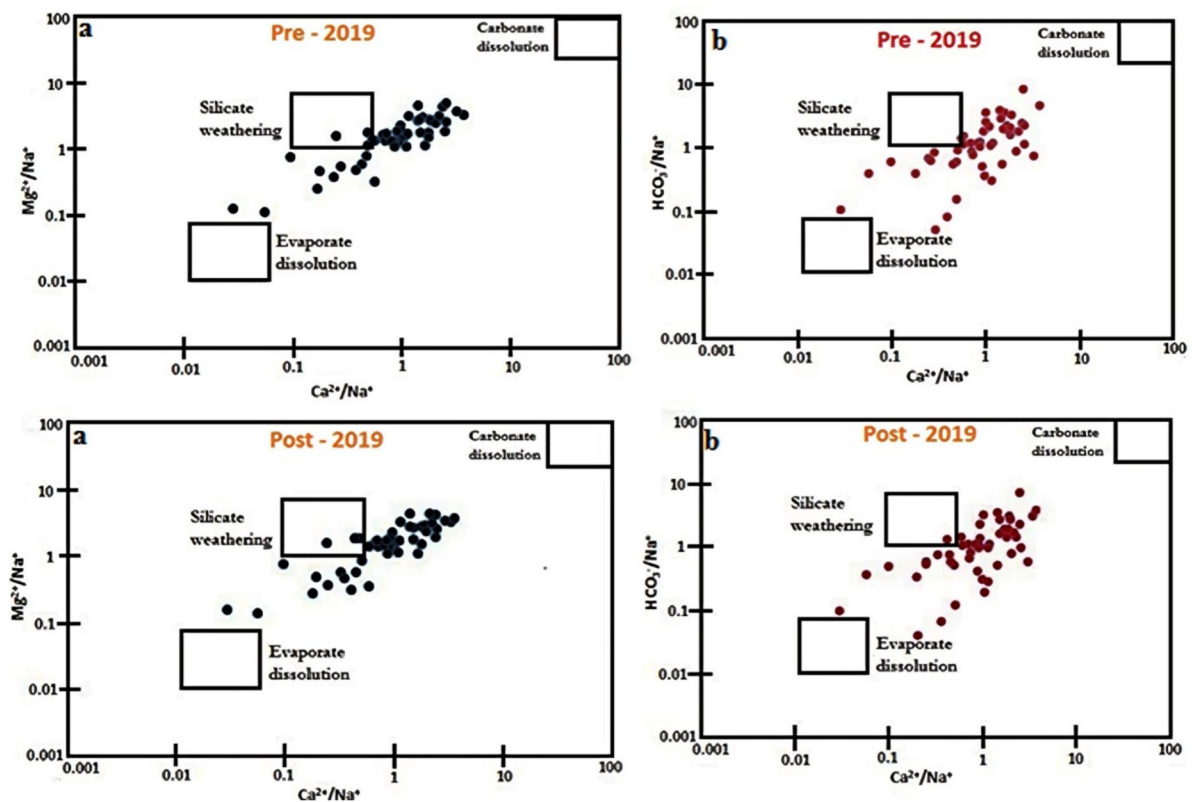
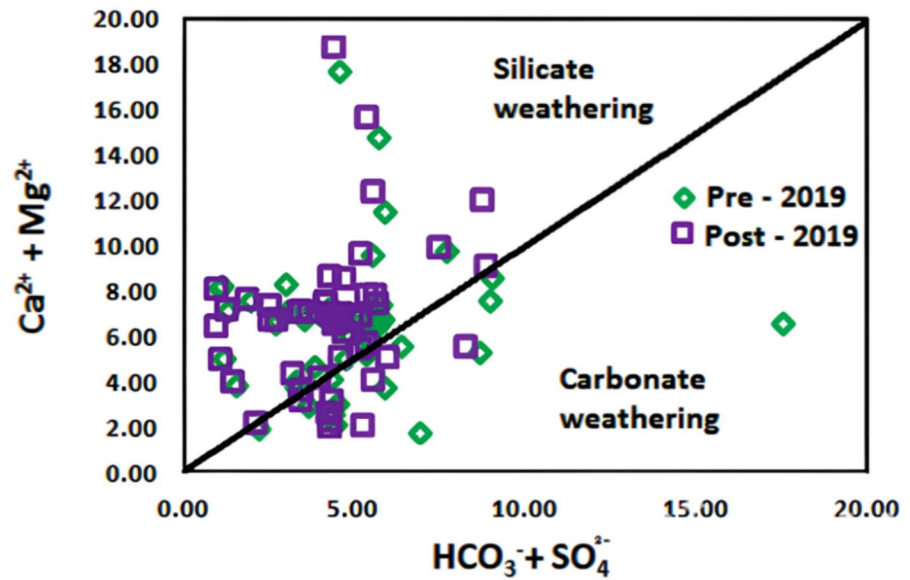


Fig. 6 Bivariate plots **a** $\text{Ca}^{2+}/\text{Na}^+$ vs $\text{Mg}^{2+}/\text{Na}^+$ **b** $\text{Ca}^{2+}/\text{Na}^+$ vs $\text{HCO}_3^-/\text{Na}^+$ in PRM and POM seasons of 2019

Fig. 7 Correlation plots a $\text{HCO}_3^- + \text{SO}_4^{2-}$ vs $\text{Ca}^{2+}/\text{Mg}^{2+}$ in PRM and POM in 2019



localized anthropogenic influence. However, evaporation dissolution appears negligible.

Overall, groundwater chemistry is primarily controlled by silicate weathering, modified secondarily by ion exchange, with limited influence from evaporation and human activities.

5 Multivariate Statistical Analysis

5.1 Correlation Matrix

Correlation analysis is widely used to examine ionic associations in groundwater and to identify common sources and controlling processes affecting water chemistry. In the present study, a correlation matrix was developed using 14 physicochemical parameters analyzed during the pre-monsoon (PRM) and post-monsoon (POM) seasons of 2019 (Fig. 8). The computed correlation coefficients are illustrated in Fig. 8. Correlation coefficients (r) greater than 0.7 were considered to represent strong relationships, while values between 0.5 and 0.7 indicate moderate correlations (Ajaykumar et al., 2021; Jabal et al., 2014).

During both PRM and POM seasons of 2019, electrical conductivity (EC) exhibited a strong positive correlation with total dissolved solids (TDS) ($r=0.91$ and 0.78 , respectively) and a moderate correlation with total hardness (TH) ($r=0.54$ and 0.51). These relationships suggest that groundwater chemistry is

influenced by natural processes such as ion exchange and leaching of carbonate minerals, in addition to contributions from sewage inputs (Wagh et al., 2016). Total dissolved solids also showed a moderate positive correlation with total hardness ($r=0.58$ in both seasons), indicating the influence of domestic and agricultural wastewater.

Magnesium (Mg^{2+}) displayed a strong positive correlation with nitrate (NO_3^-) during both PRM ($r=0.81$) and POM ($r=0.76$), highlighting the impact of anthropogenic activities such as fertilizer application and sewage infiltration. Furthermore, chloride (Cl^-) showed a moderate correlation with nitrate ($r=0.63$) during the PRM season, suggesting the percolation of wastewater and surface-derived contaminants into the groundwater system.

5.2 Factor Analysis

Varimax-rotated factor analysis (Table 4) using SPSS (version 64) was applied to interpolate the key processes shaping groundwater chemistry in the SW Y.S.R Kadapa District, where numerous inactive limestones, asbestose and barite mines remain. Fourteen variables were reduced into five significant factors explaining most of the variance (74.5% in pre-monsoon; 71.3% in post-monsoon), highlighting seasonal shifts in controlling mechanisms. Factor 1, dominated by high EC, TDS, TH, HCO_3^- in pre-monsoon and by Mg^{2+} , SO_4^{2-} , Cl^- , NO_3^- in post-monsoon,

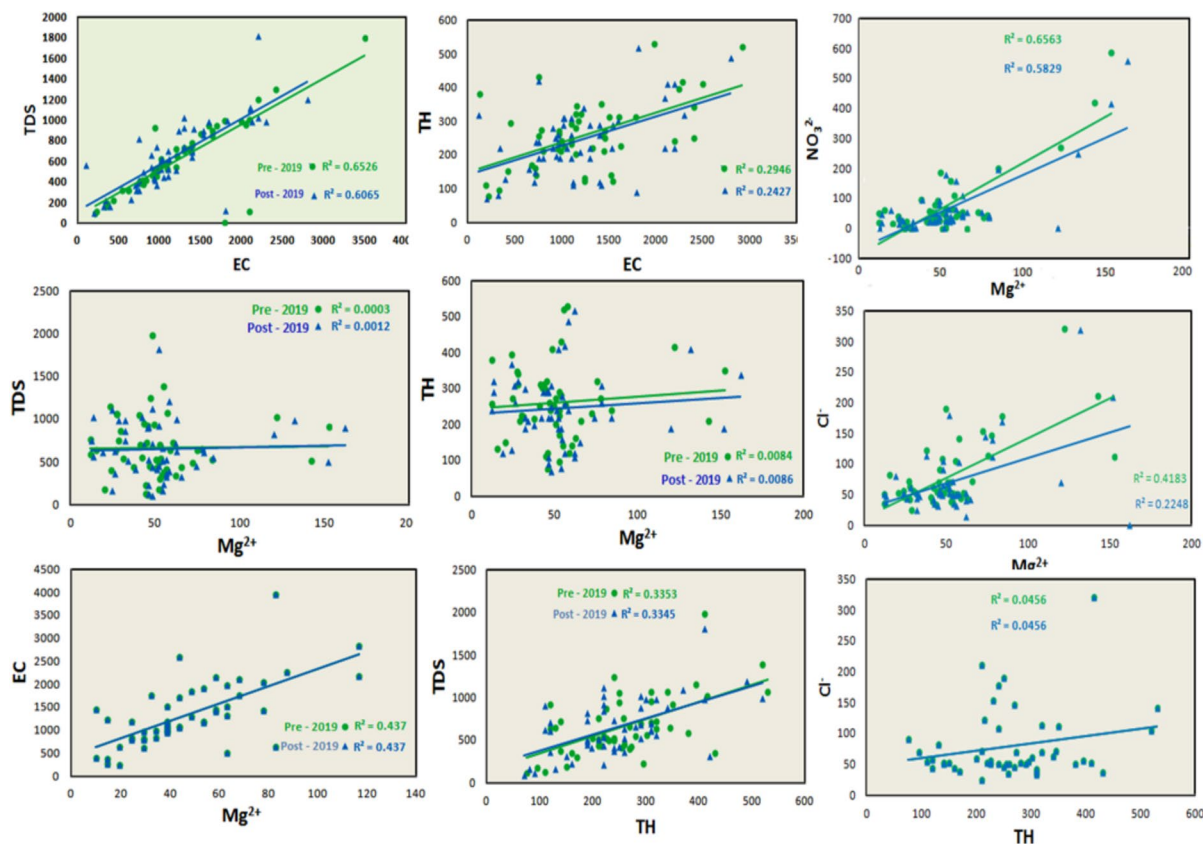


Fig. 8 Correlation plots **a** EC vs TDS **b** EC vs TH **c** Mg^{2+} vs NO_3^- **d** Mg^{2+} vs TDS **e** Mg^{2+} vs TH **f** Mg^{2+} vs Cl^- **g** Mg^{2+} vs EC **h** TH vs TDS **i** TH vs Cl

reflects combined mineral weathering and enhanced ion mobilization from recharge through weathered rock and mine waste zones. Elevated ionic strength near inactive pits suggests that recharge infiltrating exposed mine faces accelerates dissolution processes beyond background geogenic contributions.

Factor 2, mixed loadings of Mg^{2+} , Cl^- , NO_3^- , EC, and TDS indicate overlapping geogenic salinity and anthropogenic contributions; the presence of agricultural nitrate alongside salinity near mining areas suggests compounded influence of fertilizer runoff and mine-derived salts. Factor 4, dominated by K^+ and associated with nitrate and carbonate species, strongly represents fertilizer impact, but its spatial coincidence with mine clusters indicates that altered flow paths near pits and waste dumps may facilitate nitrite transport into groundwater. Factor 5, seasonal variability ($pH-SO_4^{2-}$ in PRM; Na^+ in POM) likely reflects ion exchange processes enhanced by seasonal

recharge and interaction with secondary mineral phases in fractured zones adjacent to inactive mining. Overall, the factor structure reveals that, while natural weathering processes remain basic activity, inactive mining alters subsurface flow and enhances anthropogenic signatures in groundwater chemistry.

6 Geospatial Assessment for Agro-Geogenic Contamination in Inactive Mine Impoundments

The integrated interpretation (Fig. 9) of the Aspect Map, Land Cover (2019) Map, and Groundwater Flow Direction Map provides a coherent understanding of groundwater quality deterioration in the southwestern part of the Y.S.R Kadapa District. Aspect analysis shows that the terrain is largely dominated by south-, southwest-, and southeast-facing gentle to moderate slopes, which regulate solar radiation,

Table 4 Results of the R-mode factor analysis on the chemical parameters from the study area for PRM and POM seasons

Variables	Pre-monsoon—2019					Post-monsoon – 2019				
	F1	F2	F3	F4	F5	F1	F2	F3	F4	F5
pH	−0.055	0.028	−0.006	−0.154	0.853	0.04	−0.121	0.004	0.032	0.033
EC	0.93	0.067	0.114	−0.008	0.019	0.019	0.935	−0.029	0.013	−0.045
TDS	0.943	0.061	−0.045	0.012	−0.05	0.038	0.931	−0.033	−0.117	−0.159
TH	0.651	0.082	0.141	0.333	−0.359	0.043	0.72	0.246	0.117	−0.039
Na ⁺	−0.341	0.291	0.426	−0.242	−0.048	0.179	−0.102	−0.156	0.086	0.774
K ⁺	0.128	0.219	−0.032	0.79	−0.057	0.168	0.121	0.875	0.121	−0.003
Ca ²⁺	−0.015	0.491	−0.442	0.391	0.191	0.595	−0.058	0.365	−0.379	−0.184
Mg ²⁺	−0.012	0.912	−0.002	0.155	0.047	0.876	0.047	0.134	0.086	0.122
CO ₃ ^{2−}	0.232	0.084	0.759	−0.091	0.124	0.013	0.377	−0.212	0.536	0.405
HCO ₃ [−]	0.543	0.071	−0.17	−0.485	0.255	0.084	0.242	−0.322	0.222	−0.692
SO ₄ ^{2−}	−0.011	0.016	0.144	0.444	0.533	0.587	−0.022	−0.127	−0.072	−0.023
Cl [−]	0.174	0.794	0.23	−0.191	−0.073	0.634	0.08	−0.505	0.197	0.064
NO ₃ [−]	0.062	0.898	0.04	0.232	0.025	0.793	0.11	0.31	0.132	0.141
F [−]	0.077	−0.031	−0.726	−0.252	0.01	−0.044	0.119	−0.168	−0.914	0.117
Eigen Value	2.704	2.671	1.62	1.589	1.274	2.571	2.539	1.547	1.436	1.358
% of Variance	19.315	19.08	11.574	11.353	9.1	18.367	18.132	11.05	10.259	9.698
Cumulative%	19.315	38.395	49.969	61.322	70.423	18.367	36.499	47.549	57.807	75.783

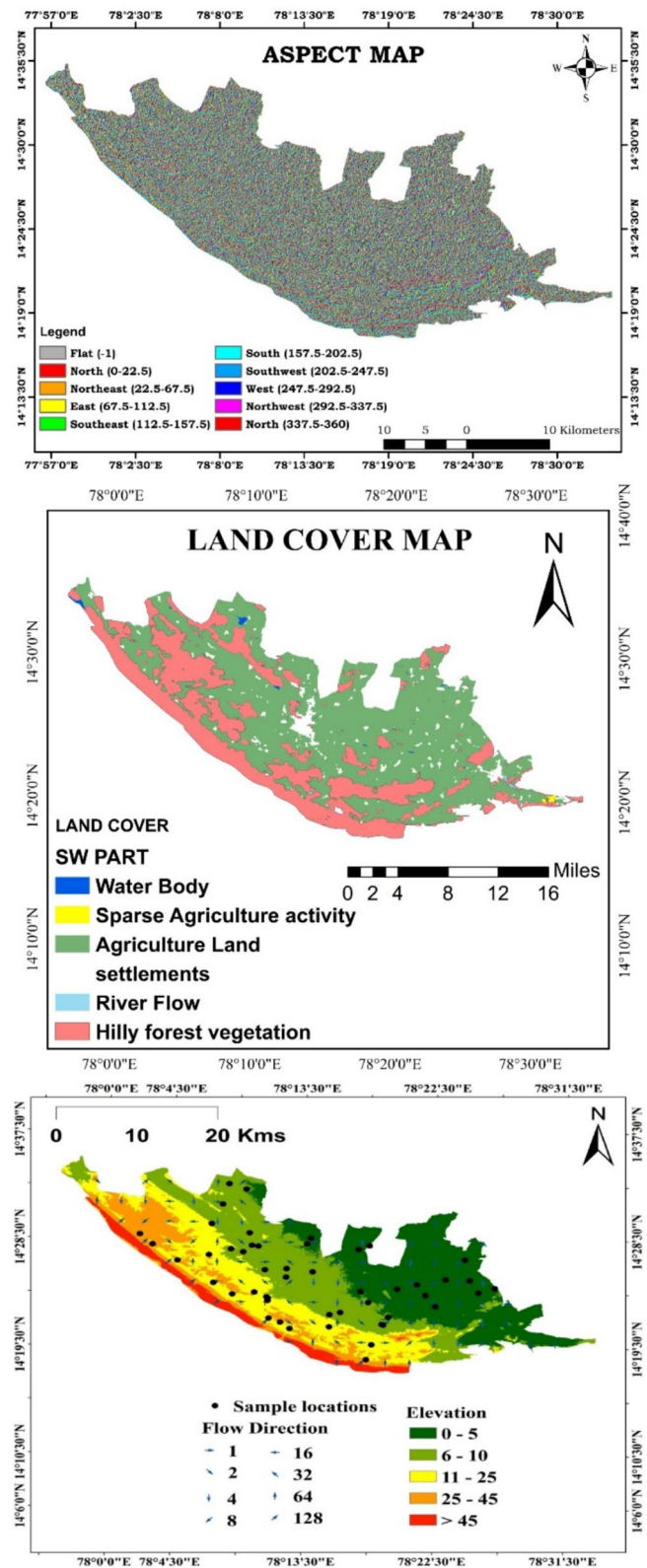
evapotranspiration, and runoff behaviour. South- and southwest-facing slopes enhance weathering of exposed carbonate formations and mine spoils, while low-lying and nearly flat sectors act as runoff convergence zones. Most inactive mine impoundments are geomorphically positioned within these depressions, enabling the accumulation of monsoonal runoff. Their location within fractured limestone and dolomite terrains promotes enhanced infiltration and strong hydraulic connectivity between surface impoundments and underlying aquifers, thereby facilitating contaminant transfer into groundwater systems.

The Land Cover (2019) map indicates that agriculture is the dominant land use, followed by hilly forest vegetation, sparse cropland, water bodies, and rural settlements. Agricultural expansion around inactive mines, particularly on low- to moderate-slope lands, is associated with intensive fertilizer application, irrigation return flows, and agrochemical usage. During rainfall, runoff enriched with nitrates (NO₃[−]), potassium (K⁺), and other dissolved constituents accumulates in Inactive mine pits, which function as recharge-enhancing depressions. In addition, domestic effluents, poultry waste, dairy farm runoff from large cattle populations, and unmanaged solid waste further increase contaminant loads. These inactive

mine reservoirs, often lacking proper closure and ecological restoration measures, act as collection points for agricultural and domestic wastes. Because many of these pits are hydraulically connected to the groundwater table, infiltrating water transports nitrate and other pollutants directly into aquifers, increasing the risk of groundwater contamination and associated health hazards.

The groundwater flow direction map reveals that, hydraulic gradients generally direct subsurface flow from southern and southwestern uplands toward central and northeastern lowlands. This downgradient movement confirms that contaminants mobilized in agro-mining landscapes migrate along natural hydraulic pathways rather than remaining localized. Elevated concentrations of fluoride (F[−]), nitrate (NO₃[−]), salinity, calcium (Ca²⁺), magnesium (Mg²⁺), and bicarbonate (HCO₃[−]) along these flow paths indicate a combined agro-geogenic contamination process. Inactive mine impoundments do not primarily generate classical acid mine drainage; instead, they modify local hydrology by enhancing recharge and intensifying water–rock interaction within fractured carbonate aquifers. This process increases dissolution of dolomite, contributing geogenic ions, while simultaneously facilitating

Fig. 9 Integrated Geo-spatial Analysis of Land Use/Land Cover (2012 & 2019), Slope, and Ground-water Flow Direction in the Southwestern Y.S.R. Kadapa District, India



the transport of anthropogenic contaminants such as nitrate and fluoride. Thus, groundwater quality deterioration in the southwestern Y.S.R Kadapa District reflects a synergistic interaction among geomorphology, land use, hydrogeology, and inactive mining structures, underscoring the urgent need for effective mine closure, waste management, and human health risk assessment in rural communities surrounding inactive mines.

7 Spatial Distribution of Hydrogeochemical Parameters Using GIS

GIS-based spatial analysis for PRM and POM 2019 indicates that groundwater quality deterioration is closely associated with areas surrounding inactive mine impoundments in the southwestern Y.S.R Kadapa District (Fig. 10). The western and north-western sectors, where abandoned mine pits and overburden dumps are concentrated, show elevated EC and TDS values in both seasons. During PRM, evaporative concentration within mine depressions increases salinity, while in POM, monsoonal recharge mobilizes accumulated salts into adjacent aquifers.

Alkaline pH conditions dominate the central and northwestern zones, reflecting enhanced carbonate dissolution within limestone and dolomitic formations. Elevated Ca^{2+} and Mg^{2+} in the southern, western, and north-central areas indicate intensified water–rock interaction promoted by prolonged residence time in inactive mine voids. Sulfate enrichment in the central (PRM) and northeastern–southwestern (POM) sectors suggest leaching from oxidized mine spoils and sulphide-bearing materials. Chloride concentration is higher in the western and northwestern regions, influenced by evaporative concentration and surface runoff accumulation in impoundments.

Nitrate contamination appears as scattered high-value patches but is pronounced near agricultural lands adjacent to mine pits, where enhanced infiltration facilitates downward migration. Fluoride enrichment in the northwestern and central zones reflects geogenic sources, further intensified by fractured, mining-disturbed aquifers. Overall, inactive mine

impoundments act as recharge concentrators and contaminant reservoirs, amplifying agro-geogenic groundwater contamination patterns.

8 Human Health Risk Assessment

In view of extensive application of fertilizers, chemicals, quality of groundwater is continuously deteriorating in recent decades due to increased awareness of health risk on humans. Health risk assessment plays a crucial role in groundwater assessment. Health assessment ensures and evaluates adverse human health effect on infants, children and adults (Wu et al., 2021; Chen et al., 2017). The present work emphasizes the assessment of health risk with respect to NO_3^- and F^- in drinking water in Southwestern part of Y.S.R Kadapa District, A.P. India. Seasonal Variation of Total Hazard Index (THI) for Adults, Children, and Infants during Pre-Monsoon and Post-Monsoon Seasons – 2019 illustrated in Table 5 and Fig. 11.

The THI results indicate that groundwater surrounding these sites is significantly impacted by contaminant migration. During PRM 2019, 90% of samples for adults and children and 76% for infants exceeded the safe limit ($\text{THI} > 1$), reflecting substantial non-carcinogenic health risk. Extremely high THI values at locations 11 (14.48), 18 (11.44), 35 (5.55), 26 (5.29), and 40 (6.92) suggest strong localized contamination, likely influenced by leachate infiltration from inactive mine pits.

In POM 2019, slight dilution effects were observed; however, 84% of samples for adults and children and 76% for infants still remained in the risk category. The persistence of high THI values (e.g., locations 11 and 18) indicates continuous pollutant loading rather than seasonal fluctuation alone.

Gentle slope conditions and proximity of agricultural lands and settlements to inactive mine impoundments facilitate percolation of nitrate-rich runoff and waste leachates into groundwater. While fluoride contamination is largely geogenic, nitrate enrichment near mine sites is primarily anthropogenic. Overall, inactive mine impoundments act as secondary contaminant reservoirs, intensifying health risks, particularly for infants.

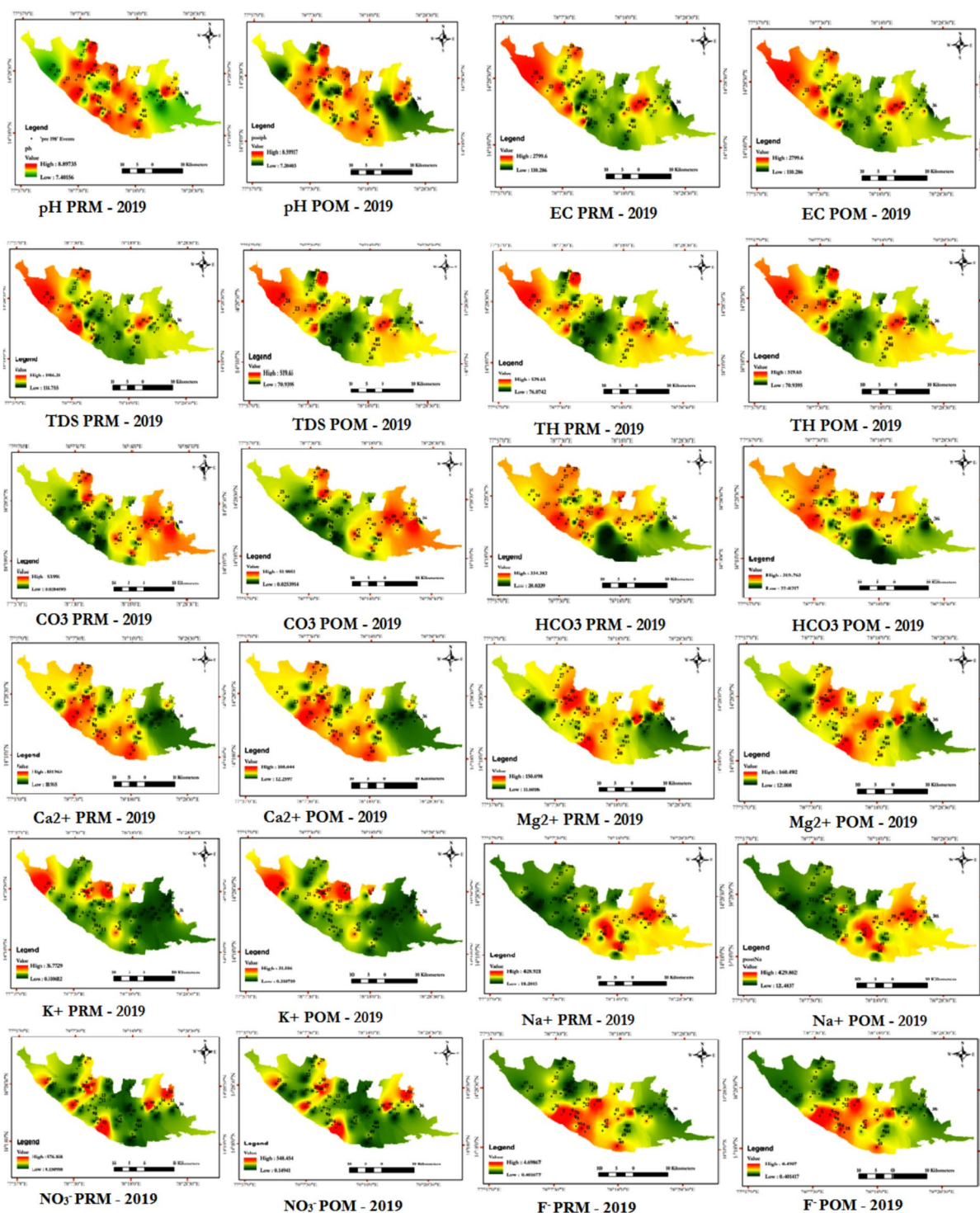


Fig. 10 Spatial distribution of various physicochemical parameters highlights their variations and potential effects on groundwater quality

Table 5 Seasonal variation of total hazard index (THI) for adults, children, and infants during pre-monsoon and post-monsoon seasons – 2019

S. No	Pre monsoon—2019			Post monsoon—2019		
	Adults	Children	Infants	Adults	Children	Infants
	THI	THI	THI	THI	THI	THI
1	1.39	1.4	1.08	1.14	1.16	1.04
2	1.13	1.15	0.88	1.08	1.1	0.84
3	1.68	1.7	1.3	1.52	1.54	1.28
4	3.73	3.78	2.89	3.55	3.6	2.85
5	3.63	3.68	2.81	3.47	3.52	2.78
6	3.12	3.16	2.42	2.94	2.98	2.38
7	4.27	4.33	3.31	4.09	4.15	3.28
8	2.7	2.73	2.09	2.52	2.56	2.06
9	1.65	1.67	1.27	1.47	1.49	1.24
10	1.49	1.51	1.15	1.46	1.48	1.13
11	11.44	11.6	8.91	11.2	11.36	8.83
12	1.94	1.96	1.5	1.82	1.85	1.46
13	2.21	2.24	1.72	2.15	2.18	1.67
14	1.67	1.69	1.3	1.61	1.63	1.25
15	0.77	0.78	0.6	0.71	0.72	0.6
16	2.05	2.08	1.59	2.03	2.05	1.57
17	1.38	1.4	1.07	1.32	1.34	1.03
18	14.48	14.68	11.29	13.78	13.98	10.75
19	1.54	1.56	1.2	1.39	1.41	1.18
20	1.62	1.65	1.26	1.45	1.47	1.22
21	1.04	1.06	0.81	1.02	1.03	0.79
22	1.14	1.15	0.88	0.8	0.81	0.87
23	1.02	1.03	0.79	0.87	0.88	0.77
24	0.77	0.78	0.6	0.52	0.52	0.6
25	4.37	4.43	3.4	4.27	4.33	3.33
26	5.29	5.37	4.12	5.11	5.18	3.98
27	2.16	2.19	1.68	2.11	2.14	1.64
28	1.01	1.02	0.78	0.96	0.97	0.74
29	3.45	3.5	2.69	3.42	3.47	2.66
30	1.42	1.44	1.11	1.37	1.39	1.07
31	3.2	3.24	2.48	3.15	3.19	2.44
32	0.87	0.88	0.67	0.86	0.87	0.66
33	2.87	2.91	2.23	2.82	2.86	2.2
34	2	2.03	1.56	1.93	1.96	1.5
35	5.55	5.62	4.32	5.39	5.47	4.2
36	1.54	1.56	1.2	1.43	1.45	1.11
37	1.2	1.22	0.93	1.19	1.21	0.92
38	1.58	1.61	1.23	1.53	1.55	1.18
39	2.67	2.71	2.07	2.6	2.63	2.01
40	6.92	7.01	5.39	6.46	6.55	5.03
41	0.91	0.92	0.7	0.91	0.92	0.7
42	1.82	1.85	1.41	1.78	1.8	1.38
43	1.2	1.21	0.93	1.14	1.15	0.88
44	0.82	0.83	0.64	0.79	0.8	0.61
45	2.72	2.76	2.11	2.68	2.72	2.08

Table 5 (continued)

S. No	Pre monsoon—2019			Post monsoon—2019		
	Adults	Children	Infants	Adults	Children	Infants
46	1.7	1.73	1.32	1.47	1.49	1.29
47	1.47	1.49	1.14	1.43	1.45	1.11
48	1.54	1.56	1.19	1.41	1.43	1.19
49	2.42	2.46	1.88	2.27	2.3	1.86
50	2.5	2.53	1.94	2.33	2.36	1.86

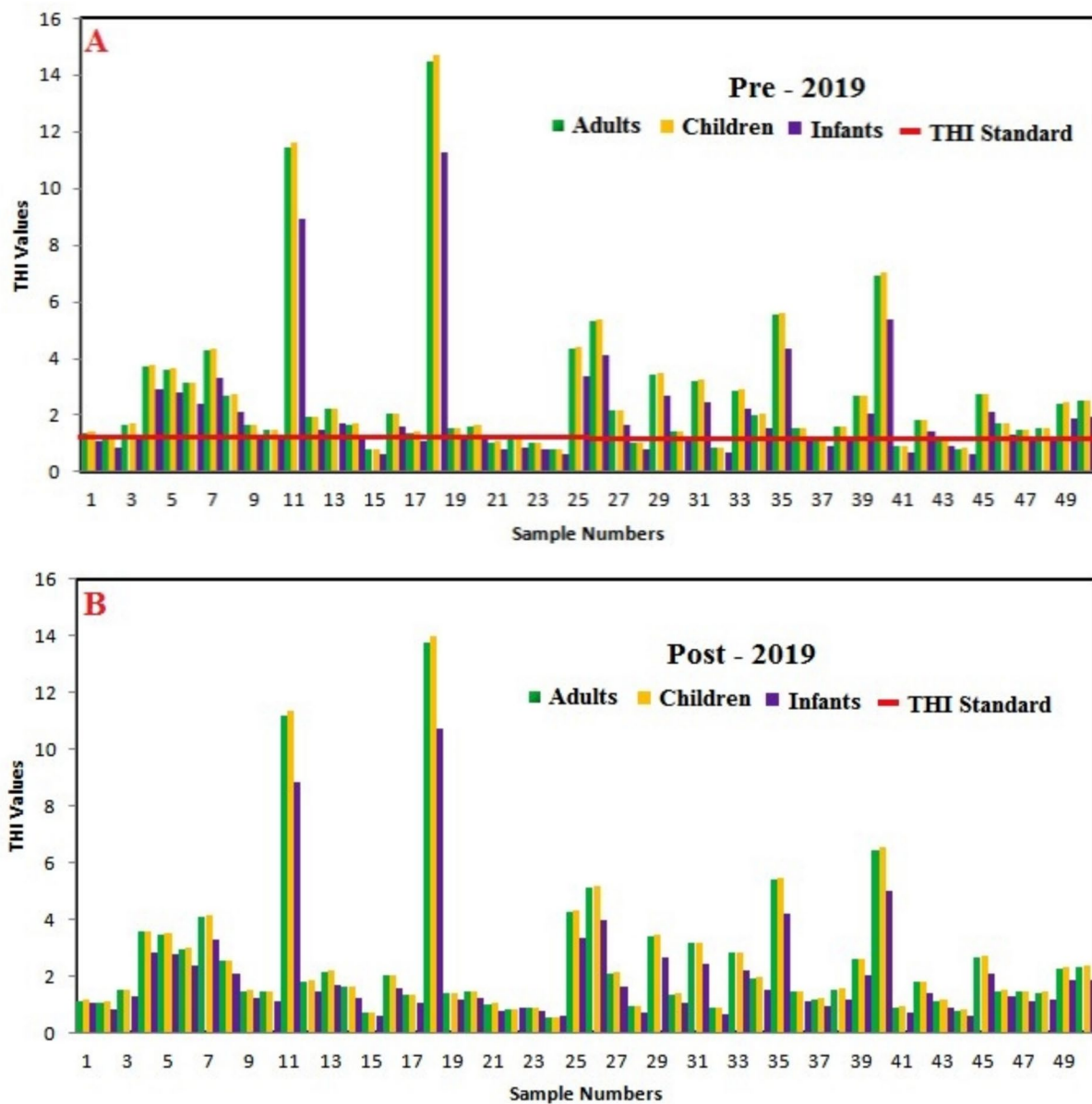


Fig. 11 Illustrates the Total Hazard Index (THI) values for adults, children, and infants across 50 groundwater sampling locations during **A** Pre-Monsoon (PRM) 2019 and **B** Post-

Monsoon (POM) 2019. The red horizontal line represents the permissible limit (THI = 1)

8.1 Heavy Metal Concentration in Groundwater

The results of heavy metal concentration & Statistical analysis of heavy metals and the comparison of concentrations of metals with World Health Organization (WHO, 2011) in groundwater of the study area were presented in Table 6.

The seasonal variation of heavy metals during PRM and POM 2019 reflects predominantly geogenic control, with limited localized influence from inactive mine impoundments. Elevated arsenic (As), nickel (Ni), lead (Pb), and strontium (Sr) concentrations are largely governed by the regional geology comprising barite, asbestos, limestone, iron ore, talc deposits, dolomitic rocks, quartzites, volcanic flows, and shale formations.

Arsenic enrichment may be related to shale horizons and minor sulfide mineral oxidation, while nickel is likely derived from weathering of mafic volcanic rocks. Lead occurrence can be attributed to natural mineralized veins and lithological weathering. The widespread exceedance of strontium strongly corresponds to dissolution of limestone and dolomitic formations, indicating dominant carbonate weathering processes.

Inactive mine impoundments may contribute locally by enhancing oxidation, infiltration, and leaching from exposed rock surfaces and waste materials; however, their impact appears spatially limited. Seasonal recharge during monsoon promotes dilution

at some locations while mobilizing trace elements at others. Overall, groundwater chemistry reflects a combined effect of lithological weathering, hydrogeochemical processes, and minor anthropogenic or mining-related influences.

9 Conclusion

The integrated seasonal hydrogeochemical and geo-spatial study carried out in the southwestern part of Y.S.R. Kadapa District shows that groundwater quality deterioration is mainly controlled by a combined natural and human-induced contamination process in mining-influenced areas. The analysis of water chemistry, slope patterns, land use, and groundwater flow indicates that inactive mine pits function as recharge zones rather than direct sources of acid mine drainage, allowing rainwater mixed with agricultural fertilizers, animal waste, and domestic sewage to infiltrate into the aquifer system. Continuous interaction between groundwater and carbonate rocks increases the concentration of calcium, magnesium, bicarbonate, and naturally occurring fluoride, while higher nitrate levels are largely linked to agricultural activities, livestock farming, and poor sanitation practices. Seasonal rainfall further enhances contaminant movement and spreads along natural groundwater flow paths, resulting in localized fluoride and nitrate hot-spots and demonstrating strong hydraulic connectivity

Table 6 Seasonal distribution of heavy metal concentrations in groundwater and comparison with WHO (2011) permissible limits – PRM and POM 2019

Heavy Metals	PRM-2019			POM—2019			Permitted values W.H.O (2011)	Exceeding permitted values	
	Minimum	Maximum	Mean	Minimum	Maximum	Mean		PRM 2019	POM 2019
As (µg/L)	0	56.3	9.1	0	60.8	6.7	10	17	11
Cd (µg/L)	0	25.6	4	0	20.5	2	30	Nil	Nil
Co (µg/L)	0	28	11	0	28.5	8.8	50	Nil	Nil
Cr (µg/L)	0	42	3.9	0	17.2	2.7	50	Nil	Nil
Fe (µg/L)	0	49	5.9	0	43	2.4	300	Nil	Nil
Li (µg/L)	0	32.3	9.7	0	30.7	8	200	Nil	Nil
Mn (µg/L)	0	40.2	3.7	0	32	2.2	400	Nil	Nil
Ni (µg/L)	0	98.3	29	0	96	28	70	9	7
Pb (µg/L)	0	92.3	11.8	0	165	127	10	8	6
Sr (µg/L)	0	1140	288.9	0	1121	285	30	46	47
Zn (µg/L)	0	2100	172	0	2087	163	3000	Nil	Nil

across the basin. Overall, the findings highlight the combined influence of geology, inactive mines, farming practices, and rural settlements on groundwater quality, posing serious health concerns such as fluorosis and nitrate-related illnesses. Future studies should emphasize long-term groundwater monitoring, seasonal assessment, and the development of sustainable management and remediation strategies, including proper mine rehabilitation, improved sanitation, and controlled fertilizer application, to ensure safe and sustainable groundwater resources in mining-affected semi-arid regions.

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Authors Contribution Sudharshan Reddy Y: Writing – original draft, Resources, Methodology, Investigation. V. Sunitha V: Writing – review & editing. Prasad M & Suvrva B: Conceptualization, review & editing. Chennakesava Reddy G: Review.

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Data Availability Data are available upon reasonable request from the authors.

Declarations

Competing Interests The authors declare no competing interests.

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